

# IOT 2025/2026 - Homework

## *Exercise 2*

Simone Amighini (person code: 10832147)

### Improved solution

In order to improve the battery lifetime of the system, we must change the period of transmission for beacon packets depending on the conditions of the patient: by using an accelerometer, the sensor is able to understand whether the patient is moving or not and, in case it is stationary, it lowers the period of transmission: we can choose to send 1 beacon every 30 minutes in this condition. It is important, yet, to keep transmitting beacons (even if at a lower rate) so that the central system is able to know which devices are effectively active. We have also to keep the sensor reactive to the start of a movement: to achieve this, we need the system to wake up using an interrupt if a movement is detected during deep sleep.

Here is the updated pseudo-code:

```
setup:
    setup_accelerometer_interrupt(action = BLE_wakeup);

loop forever:
    if motion_detected():
        send_BLE_beacon();
        sleep(10 seconds);
    else:
        send_BLE_beacon();
        deep_sleep(timeout = 30 minutes);
```

To implement the system, we need to choose small and energy efficient devices:

- Renesas DA14531 Ultra-Low Power Bluetooth 5.1 System-on-Chip
- Analog Devices ADXL362 accelerometer

### Energy saving

In the basic solution, the number of beacons sent in a day is  $\frac{86400\text{ s}}{10\text{ s}} = 8640$ .

In the proposed improvement, we consider a patient to be stationary for  $T_s\%$  of a day, so the number of beacons sent in a day is:  $86400\text{ s} \cdot \left( \frac{(1-T_s)}{10\text{ s}} + \frac{T_s}{30\text{ min} \cdot 60\text{ s/min}} \right)$ ; then, if we set  $T_s = 50\%$  as a reasonable estimation, we get that 4344 beacons are transmitted in a day.

The improvement is evident: in the considered scenario, the updated firmware sends a number of beacons equal to  $\frac{4344}{8640} = 50.28\%$  with respect to the original one.

Finally, in order to compute the energy saving, we need first to determine the energy consumption of BLE SoC related to transmission and sleep mode in both the scenarios. From the cheat sheet of the devices we get:

- energy for transmitting a beacon:  $E_{tx}=52.5 \mu J$
- sleep mode power:  $P_{sl}=5.4 \mu W$
- deep sleep mode power:  $P_{dsl}=0.81 \mu W$

Under the reasonable hypothesis that the energy consumption of the accelerometer is negligible compared to the BLE SoC, we can compute an estimation of the energy spent in a day by the device before and after the optimisation:

- $E_{original}=\#beacons_{original}\cdot E_{tx}+P_{sl}\cdot 24 h=0.9201 J$
- $E_{optimised}=\#beacons_{optimised}\cdot E_{tx}+P_{sl}\cdot 12 h+P_{dsl}\cdot 12 h=0.4963 J$

So the energy consumed by the optimised solution is estimated to be  $\frac{0.4963 J}{0.9201 J}=53.94\%$  of the energy consumed by the original one. In other terms the optimised solution consumes  $\frac{0.9201 J-0.4963 J}{0.9201 J}=46.06\%$  less energy than the original one.